

Package ‘cellpacman’

October 8, 2026

Title PAinted Cell and coMpound ANalysis (PACMAN)

Version 0.1.0

Description Concentration-response analysis of Cell Painting morphological profiles. Well-level features are z-score normalized within plate, embedded in two dimensions with UMAP, and clustered with DBSCAN separately for each known compound and jointly for unknown (unannotated) wells. A principal curve fitted through the concentration-ordered cluster centroids of each known compound defines a concentration trajectory; unknown clusters are projected onto every trajectory to identify the most similar known compound and an interpolated effective concentration. Wilcoxon rank-sum tests with multiplicity correction identify features that change along a trajectory, that discriminate unknown clusters, or that separate any two compounds or clusters. The package has no graphical dependencies beyond 'ggplot2' and 'plotly'; an interactive Shiny front end is distributed separately as the 'cellpacman-shiny' application.

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Encoding UTF-8

Depends R (>= 4.1.0)

Imports dbscan,
dplyr,
ggplot2 (>= 3.4.0),
magrittr,
plotly,
princurve,
readr,
readxl,
tibble,
tidyr,
umap,
viridisLite

Suggests ggrepel,
knitr,
rmarkdown,
testthat

VignetteBuilder knitr
Roxygen list(markdown = TRUE)
RoxygenNote 7.3.1

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compareClusterFeatures
<i>Compare features between two DBSCAN clusters</i>

Description

Tests every normalized feature for a location shift between the wells of two clusters of the same compound (or of the unknown wells), using the same Wilcoxon rank-sum test, median-difference effect size, and multiplicity correction as `compareCompoundFeatures()`.

Usage

```
compareClusterFeatures(  
  data.norm,  
  assignments,  
  first,  
  second,
```

```

    p.adjust = "BH",
    cutoff = 0.05,
    medianDiffCutoff = 0.5
  )

```

Arguments

`data.norm` A cellpainting_normalized object created by `normalize()`.

`assignments` A cluster assignment table with `WellId` and `cluster` columns, i.e. one element of `predClust$known_assignments` or `predClust$unknown_assignment` from `compoundCluster()`.

`first, second` Cluster labels to compare, as character strings ("1", "2", ...). Cluster "0" is DBSCAN noise.

`p.adjust` Multiple-testing correction passed to `stats::p.adjust()`.

`cutoff` Adjusted p-value cutoff used to call a feature significant.

`medianDiffCutoff` Absolute median-difference cutoff used to label a significant feature "Positive" or "Negative".

Value

A tibble with the same columns as `compareCompoundFeatures()`.

Examples

```

data <- loadCellPainting(exampleDataPath("features"), exampleDataPath("metadata"))
data.norm <- normalize(data, by = "Plate")
dimR <- dimReduce(data.norm, random_seed = 1)
predClust <- compoundCluster(dimR, eps = 0.5, minPts = 5)
unknown <- predClust$unknown_assignment
head(compareClusterFeatures(data.norm, unknown, "1", "2"))

```

compareCompoundFeatures

Compare features between two compound groups

Description

Tests every normalized feature for a location shift between the wells of two compounds with a two-sided Wilcoxon rank-sum test (normal approximation), reports the difference in medians (`second` minus `first`) as the effect size, and adjusts the p-values across features.

Usage

```
compareCompoundFeatures(  
  data.norm,  
  first,  
  second,  
  p.adjust = "BH",  
  cutoff = 0.05,  
  medianDiffCutoff = 0.5  
)
```

Arguments

<code>data.norm</code>	A <code>cellpainting_normalized</code> object created by <code>normalize()</code> .
<code>first, second</code>	Compound names as they appear in the Compound metadata column. NA (or the string " <code>__unknown__</code> ") denotes the pooled unannotated wells, i.e. all wells whose Compound is NA.
<code>p.adjust</code>	Multiple-testing correction passed to <code>stats::p.adjust()</code> .
<code>cutoff</code>	Adjusted p-value cutoff used to call a feature significant.
<code>medianDiffCutoff</code>	Absolute median-difference cutoff used to label a significant feature "Positive" or "Negative".

Value

A tibble with one row per feature, ordered by adjusted p-value: `feature`, `median_difference`, `p_value`, `adjusted_p_value`, and `significance` ("Positive", "Negative", "Pass p-value cutoff", or "Not significant"). Features with fewer than two finite values in either group are omitted.

See Also

`compareClusterFeatures()` for the analogous comparison between two DBSCAN clusters, `summarizeCompoundFeatures()` for the per-compound medians used to display the significant features.

Examples

```
data <- loadCellPainting(exampleDataPath("features"), exampleDataPath("metadata"))  
data.norm <- normalize(data, by = "Plate")  
head(compareCompoundFeatures(data.norm, "DMSO", "Cabo"))
```

compoundCluster	<i>Cluster compounds in reduced space</i>
-----------------	---

Description

Cluster compounds in reduced space

Usage

```
compoundCluster(dimR, method = "dbscan", eps = 0.5, minPts = 5)
```

Arguments

dimR	A cellpainting_dimred object.
method	Clustering method. Currently only "dbscan" is supported.
eps	DBSCAN epsilon.
minPts	DBSCAN minPts.

Details

Compound is required. Rows where Compound is NA are clustered together as unknown wells; non-missing values are clustered per compound.

Value

A cellpainting_clusters object.

curveEstimate	<i>Estimate per-compound concentration trajectory curves</i>
---------------	--

Description

Estimate per-compound concentration trajectory curves

Usage

```
curveEstimate(  
  dimR,  
  method = "princurve",  
  eps = 0.5,  
  minPts = 5,  
  smoother = "lowess"  
)
```

Arguments

dimR	A cellpainting_dimred object.
method	Curve-fitting method. Currently only "princurve" is supported.
eps	DBSCAN epsilon.
minPts	DBSCAN minPts.
smoother	Smoother passed to <code>princurve::principal_curve()</code> .

Value

A cellpainting_curves object.

curveProject	<i>Project unknown compound clusters onto known concentration trajectory curves</i>
--------------	---

Description

Project unknown compound clusters onto known concentration trajectory curves

Usage

```
curveProject(predCurve, newData = NULL, eps = 0.5, minPts = 5)
```

Arguments

predCurve	A cellpainting_curves object.
newData	Optional cellpainting_dimred object.
eps	DBSCAN epsilon for unknown clustering.
minPts	DBSCAN minPts for unknown clustering.

Value

A list containing all projections and per-cluster best matches.

dimReduce	<i>Run dimensionality reduction</i>
-----------	-------------------------------------

Description

Run dimensionality reduction

Usage

```
dimReduce(data.norm, method = "umap", random_seed = 1, ...)
```

Arguments

<code>data.norm</code>	A <code>cellpainting_normalized</code> object.
<code>method</code>	Dimensionality reduction method. Currently only "umap" is supported.
<code>random_seed</code>	Seed passed to the dimensionality reduction backend.
<code>...</code>	Additional arguments passed to the dimensionality reduction backend.

Details

Dimensionality reduction always computes two coordinates, stored as `x` and `y` in `result$reduced_data`.

Value

A `cellpainting_dimred` object.

<code>exampleDataPath</code>	<i>Paths to the bundled example Cell Painting tables</i>
------------------------------	--

Description

`cellpacman` ships a small Cell Painting screen in `inst/extdata`: 1,920 wells from five 384-well plates, with 664 well-level morphological features (`example_features.tsv`) and the matching plate annotation (`example_metadata.tsv`). Each plate holds three annotated compounds (`Axit` = axitinib, `Cabo` = cabozantinib, `Dacti` = dactinomycin) at eight concentrations in duplicate, 16 wells of the DMSO vehicle control, and 320 unannotated wells whose `Compound` is `NA` and which therefore enter the workflow as unknown compounds.

Usage

```
exampleDataPath(table = c("features", "metadata"))
```

Arguments

<code>table</code>	Which table to locate: "features" or "metadata".
--------------------	--

Value

A single file path to a tab-separated table that can be passed to `loadCellPainting()`.

Examples

```
exampleDataPath("features")  
exampleDataPath("metadata")
```

<code>loadCellPainting</code>	<i>Load Cell Painting feature and metadata tables</i>
-------------------------------	---

Description

Load Cell Painting feature and metadata tables

Usage

```
loadCellPainting(file.feature, file.metadata)  
  
load(file.feature, file.metadata)
```

Arguments

`file.feature` Path to a feature table or a data frame.
`file.metadata`
Path to a metadata table or a data frame.

Details

Compound is a required metadata column. NA values mark unknown compounds; non-missing values identify known compounds.

Value

A `cellpainting_data` object containing feature and metadata tables.

normalize	<i>Normalize Cell Painting features</i>
-----------	---

Description

Normalize Cell Painting features

Usage

```
normalize(data, method = "z.score", by = "Plate")
```

Arguments

data	A cellpainting_data object.
method	Normalization method. Currently only "z.score" is supported.
by	Optional grouping columns used to compute normalization per group.

Value

A cellpainting_normalized object.

plotDimred	<i>Plot reduced Cell Painting coordinates</i>
------------	---

Description

Plot reduced Cell Painting coordinates

Usage

```
plotDimred(dimR, compound_name_map = NULL)
```

Arguments

dimR	A cellpainting_dimred object.
compound_name_map	Optional named character vector used to recode compound labels.

Value

A ggplot object.

plotKnownClusters *Plot DBSCAN clusters for one known compound*

Description

Plot DBSCAN clusters for one known compound

Usage

```
plotKnownClusters(predClust, compound)
```

Arguments

predClust	A cellpainting_clusters object.
compound	Compound name to plot.

Value

A ggplot object.

plotKnownCurves *Plot a concentration trajectory curve for one known compound*

Description

Plot a concentration trajectory curve for one known compound

Usage

```
plotKnownCurves(predCurve, compound)
```

Arguments

predCurve	A cellpainting_curves object.
compound	Compound name to plot.

Value

A ggplot object.

`plotKnownCurvesPlotly`*Plot a known-compound curve as an interactive Plotly object*

Description

Plot a known-compound curve as an interactive Plotly object

Usage

```
plotKnownCurvesPlotly(predCurve, compound, plot_limits = NULL)
```

Arguments

`predCurve` A `cellpainting_curves` object.
`compound` Compound name to plot.
`plot_limits` Optional list with numeric x and y limits.

Value

A Plotly `htmlwidget`.

`plotUnknownClusters`*Plot DBSCAN clusters for unknown compounds*

Description

Plot DBSCAN clusters for unknown compounds

Usage

```
plotUnknownClusters(predClust)
```

Arguments

`predClust` A `cellpainting_clusters` object.

Value

A `ggplot` object.

plotUnknownProjections

Plot unknown-cluster projections onto one known compound curve

Description

Plot unknown-cluster projections onto one known compound curve

Usage

```
plotUnknownProjections(predCurve, projections, compound)
```

Arguments

predCurve	A cellpainting_curves object.
projections	Projection results returned by curveProject().
compound	Compound name to plot.

Value

A ggplot object.

plotUnknownProjectionsPlotly

Plot unknown-cluster projections as a native Plotly widget

Description

Plot unknown-cluster projections as a native Plotly widget

Usage

```
plotUnknownProjectionsPlotly(  
  predCurve,  
  projections,  
  compound,  
  plot_limits = NULL  
)
```

Arguments

predCurve	A cellpainting_curves object.
projections	Projection results returned by curveProject().
compound	Compound name to plot.
plot_limits	Optional list with numeric x and y limits.

Value

A Plotly htmlwidget.

readCellPainting	<i>Read a Cell Painting feature or metadata table from disk</i>
------------------	---

Description

Reads one delimited or Excel table into a tibble. The format is chosen from `ext`, which defaults to the file extension: comma-separated (`csv`), tab-separated (`tsv`, `txt`), or Excel (`xlsx`). Column names are kept exactly as stored, so feature names containing spaces (as exported by Harmony and similar software) are preserved.

Usage

```
readCellPainting(file, ext = NULL)
```

Arguments

<code>file</code>	Path to the table.
<code>ext</code>	File type used to pick the parser. Defaults to the extension of <code>file</code> ; pass it explicitly when the file name does not carry the right extension (e.g. a temporary upload).

Details

`loadCellPainting()` calls this function internally when it is given file paths; `readCellPainting()` is exported so that applications can parse an uploaded file before validation, for example to preview it.

Value

A tibble with the table contents.

Examples

```
metadata <- readCellPainting(exampleDataPath("metadata"))
metadata[, c("WellId", "Plate", "Compound", "Concentration")]
```

`selectFeatureCluster`*Select cluster-discriminating features among unknown clusters*

Description

Select cluster-discriminating features among unknown clusters

Usage

```
selectFeatureCluster(  
  predClust,  
  data.norm,  
  method = "wilcox",  
  p.adjust = "bonferroni",  
  cutoff = 0.05,  
  medianDiffCutoff = 0.5  
)
```

Arguments

<code>predClust</code>	A <code>cellpainting_clusters</code> object.
<code>data.norm</code>	A <code>cellpainting_normalized</code> object.
<code>method</code>	Statistical test. Currently only "wilcox" is supported.
<code>p.adjust</code>	Multiple-testing correction method.
<code>cutoff</code>	Adjusted p-value cutoff.
<code>medianDiffCutoff</code>	Median-difference cutoff.

Value

A tibble of pairwise cluster feature statistics.

`selectFeatureCurve` *Select curve-associated features for known compounds*

Description

Select curve-associated features for known compounds

Usage

```
selectFeatureCurve(  
  predCurve,  
  data.norm,  
  method = "wilcox",  
  p.adjust = "bonferroni",  
  cutoff = 0.05,  
  medianDiffCutoff = 0.5,  
  n_bins = 20  
)
```

Arguments

predCurve	A cellpainting_curves object.
data.norm	A cellpainting_normalized object.
method	Statistical test. Currently only "wilcox" is supported.
p.adjust	Multiple-testing correction method.
cutoff	Adjusted p-value cutoff.
medianDiffCutoff	Median-difference cutoff.
n_bins	Number of bins used for heatmap-ready summaries.

Value

A named list of per-compound feature-selection results.

`summarizeCompoundFeatures`

Summarize selected features by compound or by cluster

Description

Computes the median normalized value of each selected feature within each group, in the long format expected by heatmap functions. Unannotated wells form the "Unknown" compound group; DBSCAN noise (cluster "0") is excluded from cluster summaries.

Usage

```
summarizeCompoundFeatures(data.norm, features)  
  
summarizeClusterFeatures(data.norm, assignments, features)
```

Arguments

<code>data.norm</code>	A <code>cellpainting_normalized</code> object created by <code>normalize()</code> .
<code>features</code>	Character vector of feature names to summarize, typically the significant features returned by <code>compareCompoundFeatures()</code> or <code>compareClusterFeatures()</code> .
<code>assignments</code>	A cluster assignment table with <code>WellId</code> and <code>cluster</code> columns, i.e. one element of <code>predClust\$known_assignments</code> or <code>predClust\$unknown_assignment</code> from <code>compoundCluster()</code> .

Value

A tibble with columns `value`, `feature`, and `compound` (or `cluster`), one row per feature-group combination.

Examples

```
data <- loadCellPainting(exampleDataPath("features"), exampleDataPath("metadata"))
data.norm <- normalize(data, by = "Plate")
top <- head(compareCompoundFeatures(data.norm, "DMSO", "Cabo")$feature, 5)
summarizeCompoundFeatures(data.norm, top)
```

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